

Experiment 05: Interfacing Sensor Peripherals and Sending the Data to Cloud Platform

Roll Number: 21BIT191

Objective: Sending DHT11 Sensor Data to ThingSpeak Cloud with Raspberry Pi

Equipment and Components: Raspberry Pi – 4B, SD card/ USB Drive, Laptop, SD card reader (optional), jumper wires, DHT11 Sensor, ThingSpeak Cloud Platform Account

Theory: The DHT11 is a basic, low-cost digital sensor used to measure temperature and humidity. It operates on the principle of measuring electrical resistance, which varies with the environment's relative humidity. The sensor outputs a digital signal proportional to the temperature and humidity values.

In this experiment, the DHT11 sensor is connected to a Raspberry Pi, which serves as the processing unit. The Raspberry Pi, a compact single-board computer, is ideal for IoT applications due to its GPIO pins and compatibility with various sensors and modules.

Python is used to write a script that interfaces with the DHT11 sensor, reads the data, and processes it. The collected data is then transmitted to the ThingSpeak cloud, a platform designed for IoT analytics. ThingSpeak allows the real-time visualization and analysis of sensor data through its API, enabling remote monitoring and data-driven decision-making.

By integrating the DHT11 sensor with ThingSpeak, we can create an IoT-based environmental monitoring system. This system is capable of logging and analyzing data over time, making it useful for applications such as home automation, weather monitoring, and agricultural systems.

Interfacing with Raspberry Pi

To interface the DHT11 or DHT22 with the Raspberry Pi, three connections are required:

1. **VCC:** Connected to the 3.3V pin of the Raspberry Pi.
2. **Data:** Connected to any GPIO pin on the Raspberry Pi (commonly GPIO 4). A pull-up resistor (4.7k Ω) may be required to stabilize the signal.
3. **GND:** Connected to the ground pin of the Raspberry Pi.

By using the **Adafruit_CircuitPython_DHT** library, Python code can be written to read the temperature and humidity data from the sensor. The library simplifies the interaction with the sensor by providing easy-to-use methods to retrieve temperature and humidity values.

Python Script for Reading Data

The Python script interfaces with a DHT11 sensor connected to a Raspberry Pi to continuously monitor temperature and humidity. It uses the `adafruit_dht` library to read data from the sensor, converts the temperature from Celsius to Fahrenheit, and prints the readings. The script then sends the data to ThingSpeak, an IoT platform, via an HTTP GET request using the `requests` library. The loop repeats every 15 seconds, adhering to ThingSpeak's rate limit for free accounts. Errors are handled gracefully to ensure continuous operation.

Setting Up the ThingsSpeak Account:

Step 1: Create a ThingSpeak Account (Using an Existing MathWorks Account)

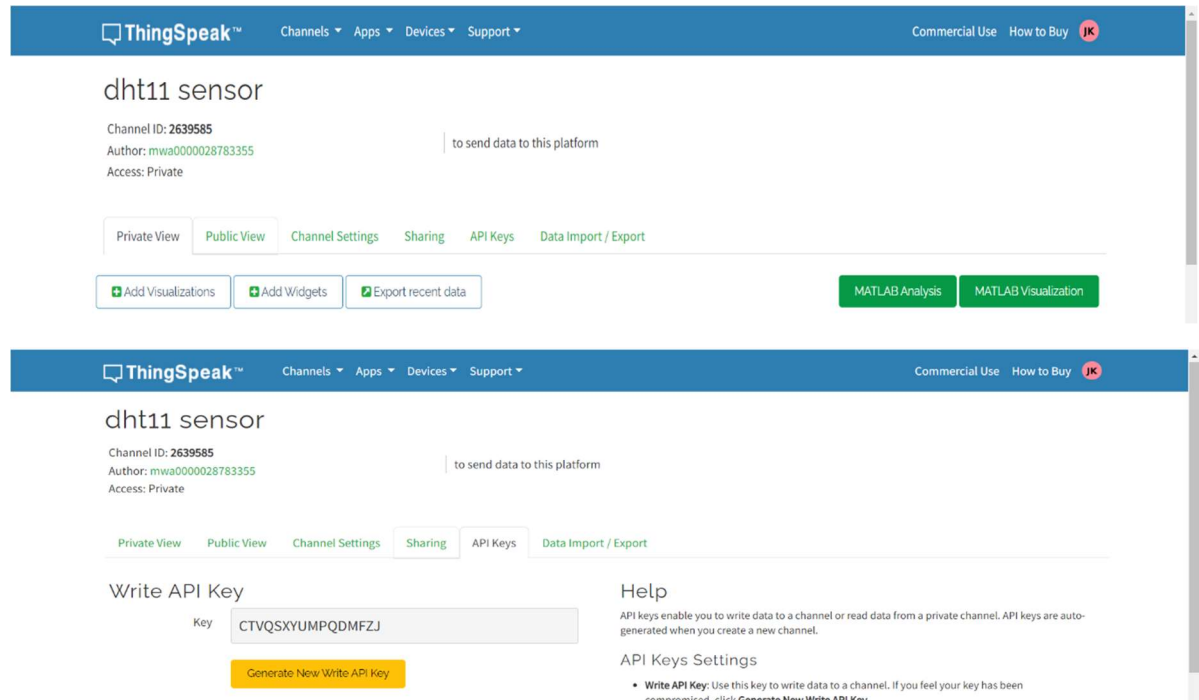
1. Visit the ThingSpeak Website:
 - Open your web browser and go to [ThingSpeak's website](https://thingspeak.com).
2. Sign In with Your MathWorks Account:
 - Click on the "Sign In" button at the top right corner of the page.
 - You'll be redirected to the MathWorks login page. Enter your MathWorks credentials (email and password).
 - After successfully logging in, you'll be redirected back to ThingSpeak, and your ThingSpeak account will be automatically created.

Step 2: Create a New Channel

1. Navigate to the "Channels" Tab:
 - Once logged in, look at the top navigation bar and click on "Channels".
2. Create a New Channel:
 - In the "Channels" tab, click on the "New Channel" button.
3. Fill in Channel Details:
 - Name: Enter a name for your channel, e.g., "Weather Station Data."
 - Description: (Optional) Enter a description, e.g., "This channel records temperature and humidity data."
 - Field 1: Enter the name "Temperature".
 - Field 2: Enter the name "Humidity".
 - Leave other fields as default unless you have specific needs.
4. Save the Channel:
 - Scroll down and click the "Save Channel" button.

Step 3: Get Your Write API Key

1. Go to the "API Keys" Tab:
 - After saving your new channel, you'll be redirected to the channel's page.
 - Navigate to the "API Keys" tab on this page.



2. Copy the Write API Key:
 - Under the "API Keys" tab, you'll see a section labeled "Write API Key".
 - Copy the Write API Key. This key will be used to send data to your ThingSpeak channel.

Procedure:

1. After setting up raspberry pi in experiment 2, now you can direct open PuTTY and repeat same steps as done previously. So, the Raspberry Pi OS desktop will appear.
2. Now, we need to update our Raspberry Pi. Run the following command to update and upgrade your Raspberry Pi to ensure that you have the latest software and libraries:

```
sudo apt update && sudo apt upgrade
```

If there are updates available, the system will prompt you to confirm. Press Y and Enter to proceed with the update. This process may take a few minutes.

3. We need to make an Virtual Environment for the Python. Navigate to your Desktop directory using the Terminal:

```
cd ~/Desktop
```

Create a new folder for your project. This folder will contain your virtual environment and the Python code:

```
mkdir dht_test
```

Move into the newly created directory:

```
cd ~/Desktop/dht_test
```

Create a virtual environment within this directory named myenv (or any other name you prefer):

```
python3 -m venv myenv
```

Activate the virtual environment:

```
source myenv/bin/activate
```

After activation, your terminal prompt will change, indicating that you are now working within the virtual environment.

```

jjkk@raspberrypi: ~/dht
File Edit Tabs Help
jjkk@raspberrypi:~/dht $ python3 -m venv myenv
jjkk@raspberrypi:~/dht $ source myenv/bin/activate
(myenv) jjkk@raspberrypi:~/dht $ python3 -m pip install adafruit-circuitpython-dht
Looking in indexes: https://pypi.org/simple, https://www.piwheels.org/simple
Collecting adafruit-circuitpython-dht
  Downloading https://www.piwheels.org/simple/adafruit-circuitpython-dht/adafruit_circuitpython_dht-4.0.4-py3-none-any.whl (7.8 kB)
Collecting Adafruit-Blinka
  Downloading https://www.piwheels.org/simple/adafruit-blinka/Adafruit_Blinka-8.47.0-py3-none-any.whl (356 kB)
    356.1/356.1 kB 388.8 kB/s eta 0:00:00
Collecting Adafruit-PlatformDetect>=3.70.1
  Downloading https://www.piwheels.org/simple/adafruit-platformdetect/Adafruit_PlatformDetect-3.73.0-py3-none-any.whl (24 kB)
Collecting Adafruit-PureIO>=1.1.7
  Downloading https://www.piwheels.org/simple/adafruit-pureio/Adafruit_PureIO-1.1.11-py3-none-any.whl (10 kB)
Collecting binho-host-adapter>=0.1.6
  Downloading https://www.piwheels.org/simple/binho-host-adapter/binho_host_adapter-0.1.6-py3-none-any.whl (10 kB)
Collecting pyftdi>=0.40.0
  Downloading https://www.piwheels.org/simple/pyftdi/pyftdi-0.55.4-py3-none-any.whl (145 kB)

```

- Now, we need to install the library in the virtual environment. Install the Adafruit CircuitPython DHT Library.

With the virtual environment active, install the Adafruit CircuitPython DHT library by running the following command:

```
python3 -m pip install adafruit-circuitpython-dht
```

This library will allow your Raspberry Pi to interface with the DHT11/DHT22 sensor and retrieve temperature and humidity readings. Also we need to download the request library in the virtual environment so install it by form terminal of myenv using:

```
pip install requests
```

5. Then Write the Python Script. In the same directory where you created the virtual environment (`~/Desktop/dht_test`), create a new Python file, for example, `dht_basic.py`. Open the file in a text editor and copy the following Python code into it:

Save the file and close the editor.

6. Further, Run the Python Script. Ensure that the virtual environment is active. If it's not, reactivate it by running: `source myenv/bin/activate`

```

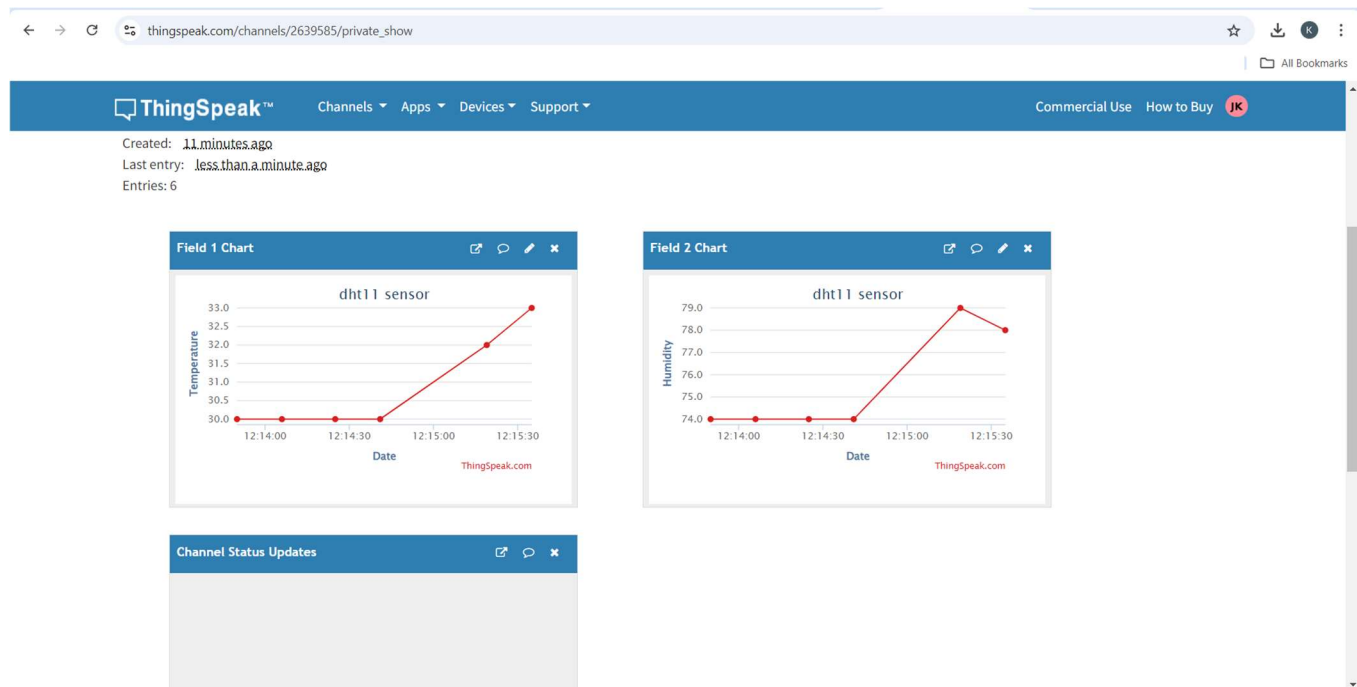
4
5 import time
6 import board
7 import adafruit_dht
8 import requests
9
10 # Sensor data pin is connected to GPIO 4
11 sensor = adafruit_dht.DHT11(board.D4)
12
13 # Your ThingSpeak Channel ID and Write API Key
14 THINGSPEAK_CHANNEL_ID = 2639585
15 THINGSPEAK_API_KEY = "CTVQ5XYUMPQDMFZJ"
16 THINGSPEAK_URL = "https://api.thingspeak.com/update"
17
18 while True:
19     try:
20         # Read the sensor data
21         temperature_c = sensor.temperature
22         temperature_f = temperature_c * (9 / 5) + 32
23         humidity = sensor.humidity
24         print("Temp={0:0.1f}C, Temp={1:0.1f}F, Humidity={2:0.1f}%".format(temperature_c, temperature_f, humidity))
25
26         # Send data to ThingSpeak
27         response = requests.get(THINGSPEAK_URL, params={
28             'api_key': THINGSPEAK_API_KEY,
29             'field1': temperature_c,
30             'field2': humidity
31         })
32
33         if response.status_code == 200:
34             print("Data successfully sent to ThingSpeak.")
35         else:
36             print(f"Failed to send data to ThingSpeak. Status code: {response.status_code}")
37
38     except RuntimeError as error:
39         # Errors happen fairly often with DHT sensors, just keep going
40         print(error.args[0])
41         time.sleep(2.0)
42         continue
43     except Exception as error:
44         sensor.exit()
45         raise error
46
47     time.sleep(15.0) # ThingSpeak rate limit for free accounts

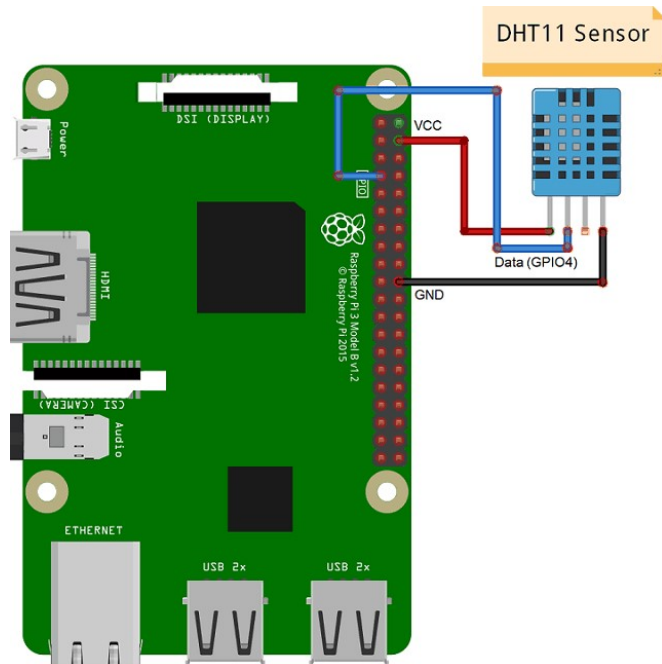
```

Run the Python script using the following command: `python dht_basic_cloud.py`. After the Successful interpretation of the Code it will prompt as the Data is successfully sent to ThingSpeak. After 15 seconds the data will be visible in the ThingSpeak Platform.

```
jjkk@raspberrypi: ~/dht/myenv
File Edit Tabs Help
jjkk@raspberrypi:~/dht $ source myenv/bin/activate
(myenv) jjkk@raspberrypi:~/dht $ ls
myenv
(myenv) jjkk@raspberrypi:~/dht $ cd myenv
(myenv) jjkk@raspberrypi:~/dht/myenv $ ls
bin cloud.py dhts.py include lib lib64 pyvenv.cfg
(myenv) jjkk@raspberrypi:~/dht/myenv $ python cloud.py
Temp=30.0°C, Humidity=74.0%
Data successfully sent to ThingSpeak.
Temp=30.0°C, Humidity=74.0%
Data successfully sent to ThingSpeak.
```

Result:



Block Diagram:

Discussion: This experiment demonstrates the integration of a DHT11 sensor with a Raspberry Pi for real-time environmental monitoring, leveraging IoT technology. The sensor readings for temperature and humidity are sent to the ThingSpeak cloud, where data can be visualized and analyzed remotely. This setup is particularly useful for applications such as home automation, agriculture, and weather stations.

Conclusion: The experiment successfully demonstrates how to interface a DHT11 sensor with a Raspberry Pi and send data to the ThingSpeak cloud for real-time monitoring. This experiment highlights the ease of implementing IoT solutions for environmental data collection and analysis. The system is robust, capable of handling errors and efficiently logs data at regular intervals, making it a practical solution for various IoT applications.